Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(An Empowered Autonomous Institute Affiliated to University of Mumbai)

Department of Electronics & Telecommunication Engineering

EXPERIMENT NO. 4

Design, Configuration and Verification of LAN/WAN Connectivity using Cisco Packet Tracer

Name	Chinmay Tawade	**UID**	2024200132
Batch	B4	**Experiment**	4

1. Objective

This experiment is aimed at building hands-on familiarity with logical network design and Layer-3 configuration inside Cisco Packet Tracer. By the end of the exercise, the following should be achieved:

- Construct a simple point-to-point network joining two end hosts through an intermediate switch and router.
- Construct a two-LAN network in which each LAN sits behind its own router, with the routers joined over a serial WAN link.
- Assign static IPv4 addressing (host address, subnet mask, default gateway) to every PC and router interface involved.
- Configure static routing on both routers so that traffic can cross from one LAN to the other.
- Verify, using both the command-line ping utility and Packet Tracer's Simulation mode, that the network is non-functional before configuration and fully functional afterwards.

2. Software / Tools Used

Tool	Purpose
Cisco Packet Tracer 9.0.1	Logical network design, device configuration and packet-level simulation
Router / Switch CLI (IOS)	Interface addressing, clock rate and static route configuration

3. Theoretical Background

A computer network is only as useful as the addressing and routing information carried on top of the physical wiring. Cisco Packet Tracer is useful precisely because it separates these two concerns visually: the Logical workspace shows cables, link lights and device icons, while a device's Config/CLI tab shows the addressing and routing state that actually decides whether a packet gets delivered.

Every end device needs three pieces of Layer-3 information before it can talk beyond its own segment: a host IP address, a subnet mask that defines the boundary of its local network, and a default gateway, which is simply the router interface that hands off any traffic destined outside that boundary. A switch, by contrast, needs none of this to forward frames within a single LAN – it works at Layer 2, learning MAC addresses on the fly. A router, however, must have every one of its interfaces addressed, brought up with no shutdown, and – in a network with more than one subnet – must know an explicit or learned route to any subnet it does not directly own. Missing any one of these leaves the topology physically cabled but logically "down".

The ICMP ping utility is the standard way to test this end-to-end. When a ping is issued, the source host encapsulates an ICMP Echo Request inside an IP packet addressed to the destination and hands it to its default gateway if the destination is off-segment. Packet Tracer's Simulation mode makes this process visible: each hop the packet takes – switch, router, router, switch – is logged with a timestamp, and the Last Status column reports Successful only once an Echo Reply has traced its way all the way back to the source. A Failed status at any stage of the experiment is therefore not a fault in the software; it is a direct, observable consequence of incomplete Layer-3 configuration, and resolving it is the actual learning objective of this exercise.

Two connection patterns are explored below: a point-to-point arrangement, where a single switch and router sit between two hosts, and an inter-LAN arrangement, where two independent LANs – each with its own switch and router – are joined by a serial WAN link and reach each other only via static routes configured on both routers.

4. Experiment Design Overview

Part	Topology	Devices Used	Goal
A	Point-to-Point (single LAN)	2 PCs, 1 switch, 1 router	Show down → configured → up transition on a single segment
B	Bus / Inter-LAN over WAN	4 PCs, 2 switches, 2 routers	Route traffic between two independent LANs over a serial link

5. Part A – Point-to-Point Network

5.1 Topology

PC-A is connected to Switch0, which in turn connects to Router0, which connects onward to PC-B. This gives the simplest possible multi-hop path and is a good baseline for observing the down-to-up transition before adding a second LAN.

5.2 Addressing Plan

Device / Interface	IP Address	Subnet Mask	Default Gateway
PC-A	192.168.1.3	255.255.255.0	192.168.1.1
PC-B	192.168.0.3	255.255.255.0	192.168.1.1
Router0 (LAN-facing interface)	192.168.1.1	255.255.255.0	192.168.1.1

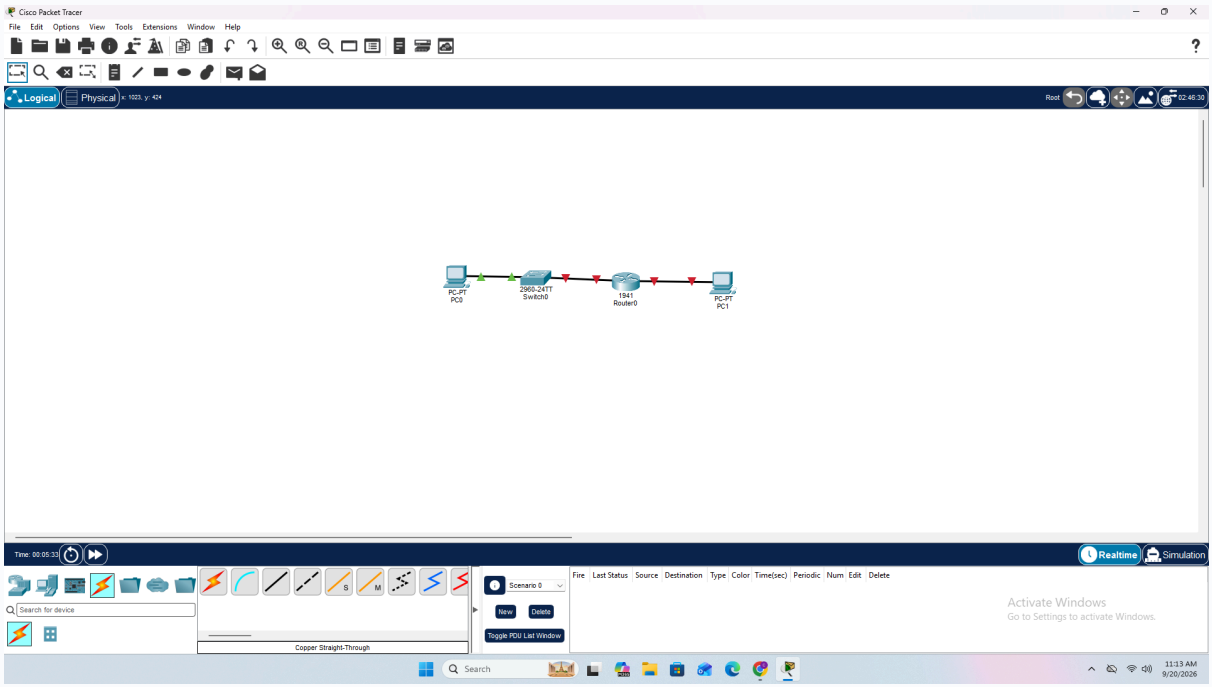
Fill in the addressing scheme actually used in your Packet Tracer file before submission.

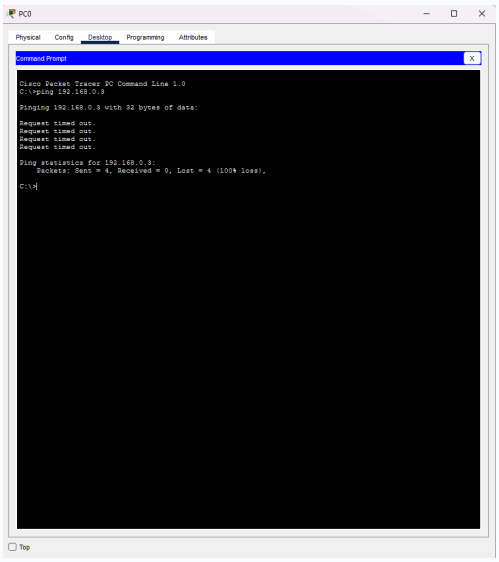
5.3 Procedure

1. Place PC-A, PC-B, one 2960-24TT switch and one 1941 router on the Logical workspace and cable them PC-A – Switch0 – Router0 – PC-B.
2. Before configuring anything, switch to Simulation mode and send an ICMP ping from PC-A to PC-B to confirm the network is "down".
3. Open PC-A's Desktop → IP Configuration and assign a static IP address, subnet mask and default gateway; repeat for PC-B.
4. Open Router0's CLI, enter the LAN-facing interface, assign its IP address, and issue no shutdown.
5. Re-run the ping from PC-A to PC-B in Simulation mode and observe the packet moving hop-by-hop.
6. Confirm final connectivity from the PC-A command prompt using ping <PC-B IP address>.

5.4 Observation 1 – Network Status Before Configuration

With cabling complete but no addressing applied, the ping from PC-A to PC-B should be captured as Failed in the Simulation Panel.



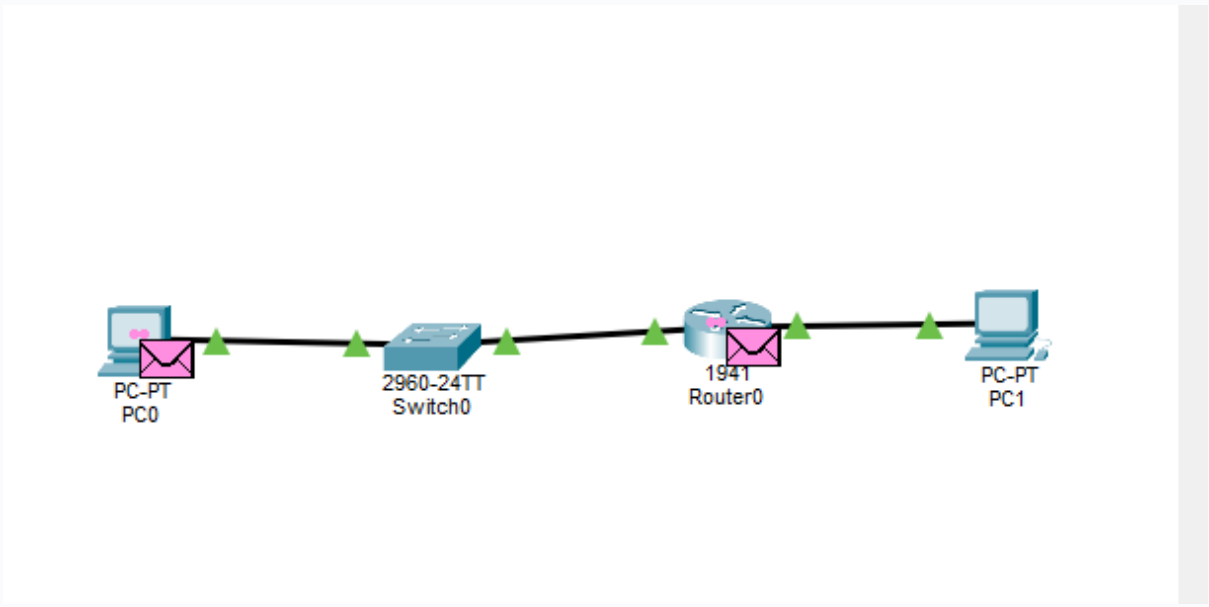


Topology view with ping result = Failed

Fig. A.1 – Point-to-point topology before configuration (ping: Failed)

5.5 Observation 2 – Devices Configured, Packet In Transit

While the re-run ping is being traced, capture a frame where the Simulation Panel shows the packet In Progress at an intermediate device.

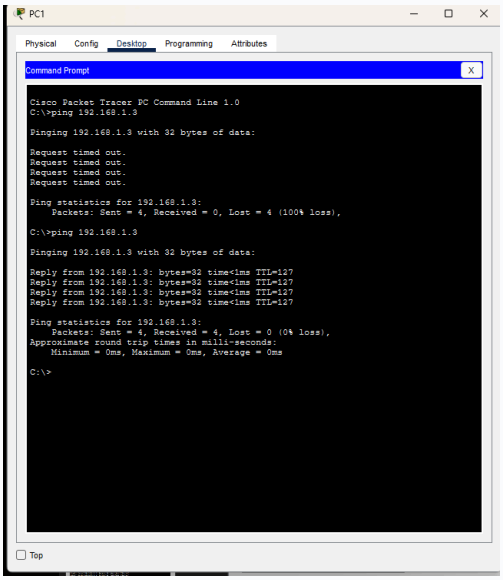


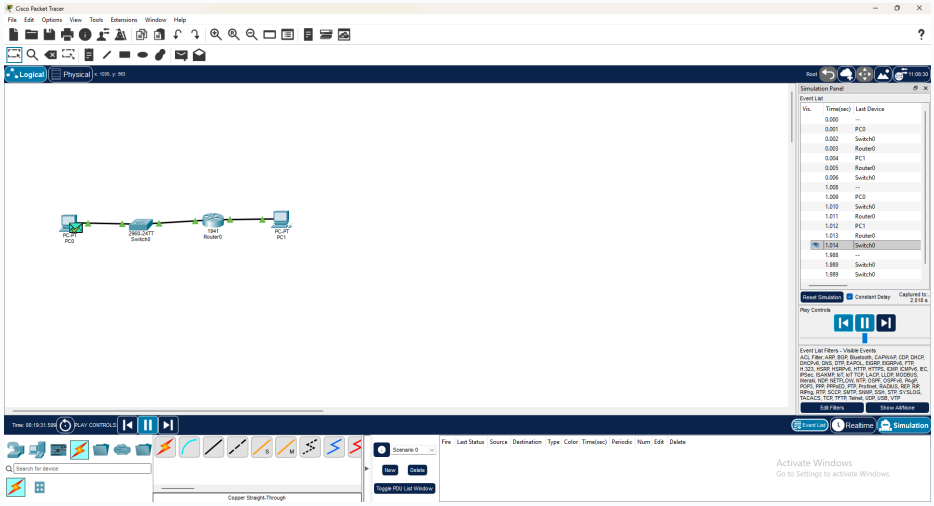
Simulation Panel with packet status = In Progress

Fig. A.2 – Packet in transit through Router0 after configuration

5.6 Observation 3 – Network Status After Configuration

Once every device is addressed, the same ping should now trace all the way to PC-B and back with a Last Status of Successful.





Full event list with Last Status = Successful

Fig. A.3 – Point-to-point topology after configuration (ping: Successful)

5.7 Result

The point-to-point network moved from a non-functional to a fully functional state purely as a result of Layer-3 configuration – the physical cabling never changed between the two states.

6. Part B – Inter-LAN Network over a WAN Link

6.1 Topology

Two independent LANs are built. LAN1 hosts PC-A and PC-B behind Switch0 and Router0; LAN2 hosts PC-C and PC-D behind Switch1 and Router1. Router0 and Router1 are joined directly by a serial cable, forming the WAN backbone that the two LANs must route across to reach each other.

6.2 Addressing Plan

Device / Interface	IP Address	Subnet Mask	Default Gateway
PC-A	192.168.1.2	255.255.255.0	192.168.1.1
PC-B	192.168.2.2	255.255.255.0	192.168.2.1
PC-C	192.168.3.2	255.255.255.0	192.168.3.1
PC-D	192.168.4.2	255.255.255.0	192.168.4.1
Router0 LAN interface	192.168.1.1	255.255.255.0	192.168.1.1
Router0 Serial interface	10.0.0.1	255.255.255.252	–
Router1 LAN interface	192.168.3.1	255.255.255.0	–
Router1 Serial interface	10.0.0.2	255.255.255.252	–

Fill in the addressing scheme, and the static routes configured on each router, actually used in your Packet Tracer file.

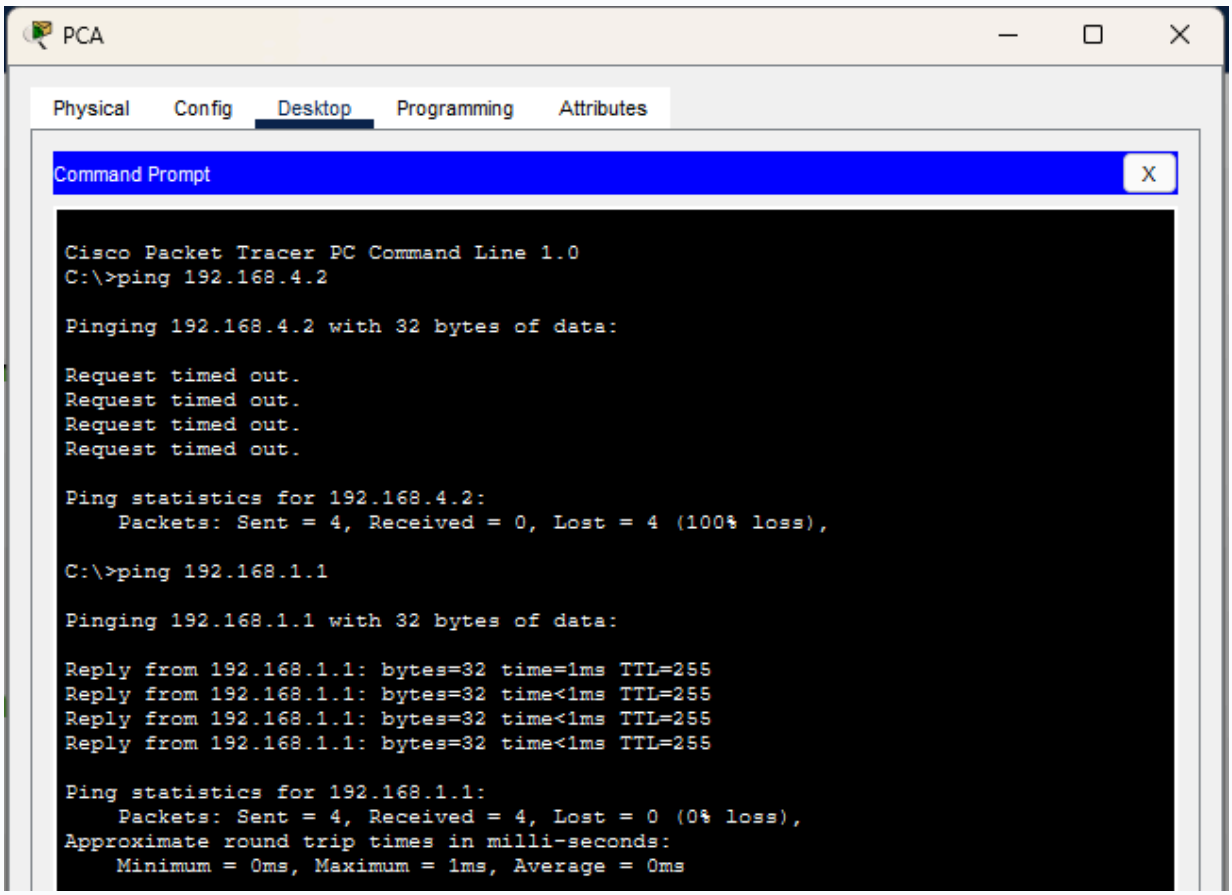
6.3 Procedure

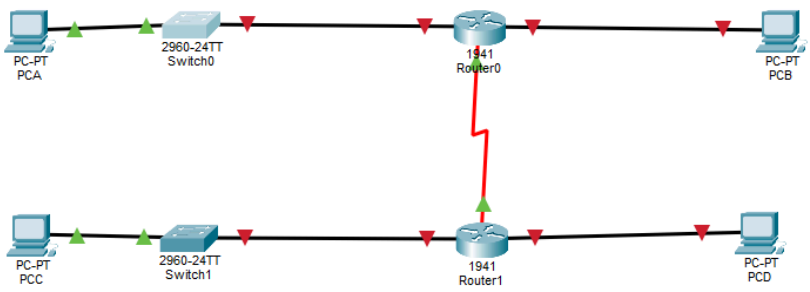
1. Build LAN1 (PC-A, PC-B, Switch0, Router0) and LAN2 (PC-C, PC-D, Switch1, Router1) as separate segments, then connect Router0 and Router1 with a serial cable to form the WAN link.
2. Before any configuration, send a ping from PC-A to PC-D in Simulation mode and confirm it is Failed – the physical topology is complete but Layer-3 information is absent.
3. Assign static IP addressing and default gateways to all four PCs via Desktop → IP Configuration.
4. On each router, add a HWIC-2T serial module in the Physical view if the serial interface is not already present.
5. On Router0's CLI: address the LAN interface and the serial interface, set a clock rate on the DCE end, issue no shutdown on both, and add static routes to the remote LAN2 subnet(s).

6. On Router1's CLI: address the LAN interface and the serial interface, issue no shutdown, and add static routes back to LAN1 subnet(s).
7. Re-run the ping from PC-A to PC-D in Simulation mode and trace it hop-by-hop across both routers.
8. Confirm connectivity from the PC-A command prompt using ping <PC-D IP address> and check for 0% packet loss.

6.4 Observation 1 – Network Status Before Configuration

Capture the topology with the ping from PC-A to PC-D marked Failed, before any router or PC has been addressed.



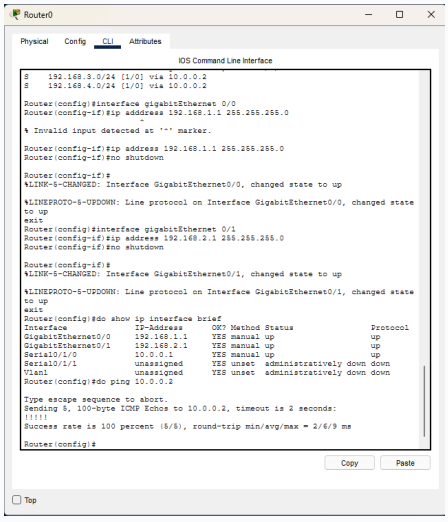


Topology view with ping PC-A → PC-D = Failed

Fig. B.1 – Inter-LAN topology before configuration

6.5 Observation 2 – Router Configuration (CLI)

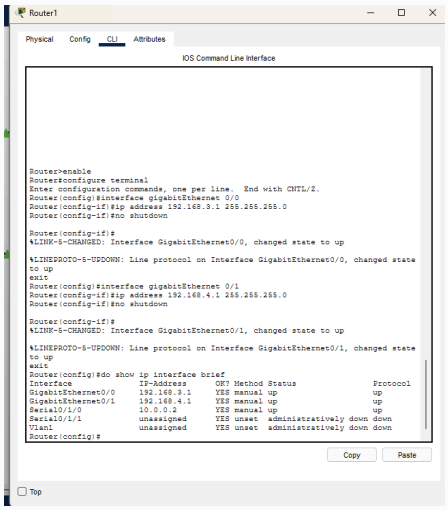
Capture the CLI output on Router0 showing the serial interface being addressed, the clock rate being set and the static routes to LAN2 being added.



Router0 CLI – interface, clock rate and ip route commands

Fig. B.2 – Router0 CLI configuration

Repeat for Router1, showing its serial addressing and the static routes back to LAN1.

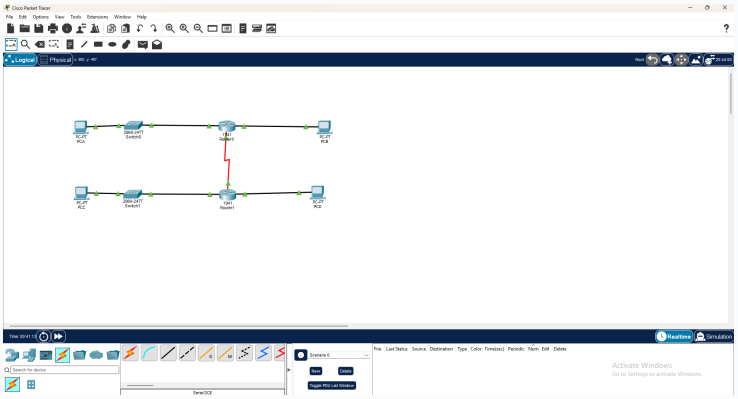


Router1 CLI – interface and ip route commands

Fig. B.3 – Router1 CLI configuration

6.6 Observation 3 – Simulation Mode, Packet In Transit

Capture at least one frame of the Simulation Panel showing the ICMP packet crossing the serial WAN link between the two routers.



Simulation Panel – packet crossing the serial link

Fig. B.4 – Packet in transit across the Router0 – Router1 WAN link

6.7 Observation 4 – Network Status After Configuration

Capture the completed event list with a Last Status of Successful for the PC-A to PC-D ping, and the command-prompt ping result confirming 0% loss.

```
Router(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

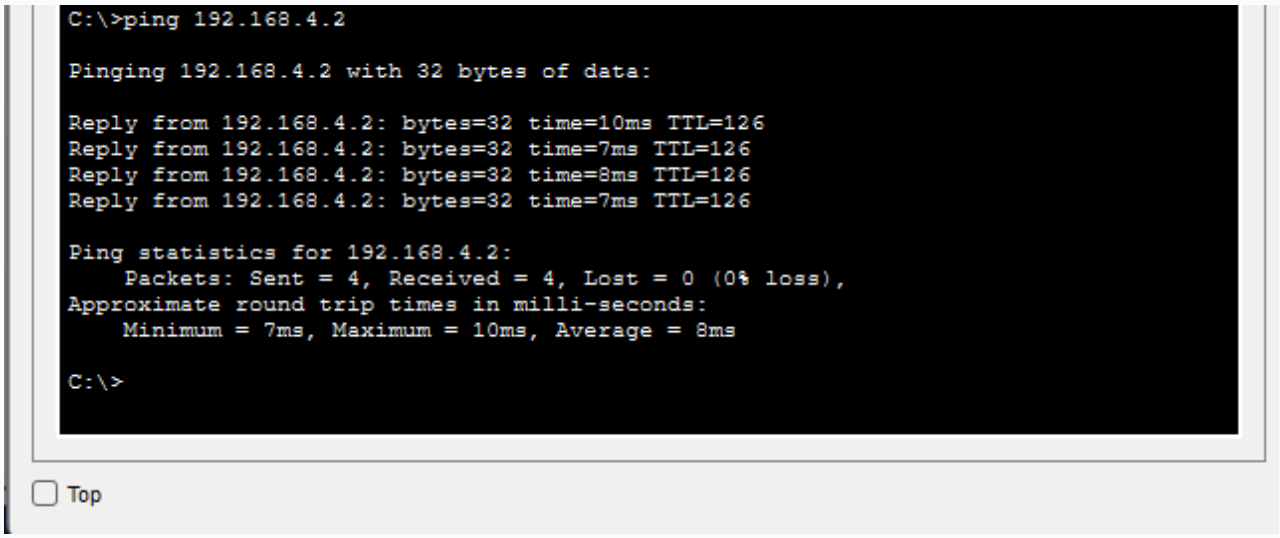
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/1/0
L       10.0.0.1/32 is directly connected, Serial0/1/0
S    192.168.3.0/24 [1/0] via 10.0.0.2
S    192.168.4.0/24 [1/0] via 10.0.0.2

Router(config)#
```

Copy Paste

Event list with Last Status = Successful

Fig. B.5 – Inter-LAN topology after configuration (ping: Successful)



PC-A command prompt ping output, 0% loss

Fig. B.6 – Command-prompt verification of PC-A → PC-D connectivity

6.8 Result

End-to-end connectivity between LAN1 and LAN2 was established only after both routers carried correct interface addressing and static routes to each other's subnet, confirming that inter-LAN reachability depends on routing information and not on the WAN link being physically up alone.

7. Results Summary

Part	Before Configuration	After Configuration
A – Point-to-Point	Ping PC-A → PC-B: Failed	Ping PC-A → PC-B: Successful
B – Inter-LAN / WAN	Ping PC-A → PC-D: Failed	Ping PC-A → PC-D: Successful, 0% loss

8. Conclusion

Both parts of this experiment reached the same underlying conclusion through different topologies. In the point-to-point network, a single switch and router were enough to demonstrate that physical connectivity and logical reachability are independent conditions – the link lights were green from the moment the cables were placed, yet packets could not be delivered until every device carried valid addressing. In the inter-LAN network, this idea extended naturally to routing: two correctly addressed LANs still could not reach each other until both routers were told, via static routes, how to find the subnet on the far side of the WAN link.

Working through the failure state deliberately – observing the Failed ping before touching any configuration – made the effect of each subsequent step easier to isolate: assigning host addressing fixed local delivery, while assigning router interfaces and static routes fixed delivery across the WAN link. Packet Tracer's Simulation mode was particularly useful here, since it made the packet's hop-by-hop path a directly observable quantity rather than something inferred only from a pass/fail ping result. Overall, the exercise reinforced that a working network is the product of three layers acting together – physical cabling, host/interface addressing, and routing information – and that Packet Tracer's ability to freeze and inspect any one of them makes it a practical tool for troubleshooting Layer-3 issues before ever touching real hardware.